



In [28]: *#Slicing*

```
string = input("enter string")
print(string)
print(string[5:10])
print(string[0:5])
print(string[7:])
print(string[:5])
print(string[-7:-1])
```

vedanti  
ti  
vedan

vedan  
vedant

In [27]: *#casting*

```
n= int(input("enter no"))
a=float(n)
print(a)
a=str(n)
print(a)
```

55.0  
55

In [51]: *#string Functions*

```
string = input("enter string")
print(string)
print(len(string))
print(string.upper())
print(string.lower())
print(string.capitalize())
print(string.replace("a","n"))
print(string.split())
print(string.swapcase())
print(string.title())
print(string.casefold()) #covert in lowercase
print(string.count('v'))
```

bhavi  
5  
BHAVI  
bhavi  
Bhavi  
bhnvi  
['bhavi']  
BHAVI  
Bhavi  
bhavi  
1

In [61]: *#Types of operators:- arithmetic operator*

```
a= int(input("enter first no="))
b= int(input("enter second no="))
print(a+b)
print(a-b)
print(a*b)
print(a/b)
print(a%b)
print(a//b)
print(a**b)
```

```
6
0
9
1.0
0
1
27
```

In [69]: *#assignment operator*

```
a= int(input("enter first no="))
b= int(input("enter second no="))
x=a
print("initial valuee of x:",x)

x+=b
print("after x+=b:",x)

x-=b
print("after x-=b:",x)

x*=b
print("after x*=b:",x)

x/=b
print("after x/=b:",x)
```

```
initial valuee of x: 2
after x+=b: 4
after x-=b: 2
after x*=b: 4
after x/=b: 2.0
```

In [3]: *#tuple*

```
t= tuple(map(int,input("enter elements seperated by space : ").split()))

print("tuple:",t)
print("1.length:",len(t))
print("2.maximum:",max(t))
print("3.minimum:",min(t))
print("4.sum:",sum(t))
```

```

print("5.count of 3:",t.count(3))
print("6.index of element:",t.index(t[3]))
print("7.sorted:",tuple(sorted(t)))
print("8.slicing:",t[:2])
print("9.repetation:",t*2)
print("10.membership:",5 in t)

```

```

tuple: (2, 3, 4, 5, 4, 3, 2, 4)
1.length: 8
2.maximum: 5
3.minimum: 2
4.sum: 27
5.count of 3: 2
6.index of element: 3
7.sorted: (2, 2, 3, 3, 4, 4, 4, 5)
8.slicing: (2, 3)
9.repetation: (2, 3, 4, 5, 4, 3, 2, 4, 2, 3, 4, 5, 4, 3, 2, 4)
10.membership: True

```

In [1]: `#list`

```

lst= (input("enter elements seperated by space: ").split())
print("List:",lst)
print("1.length:",len(lst))
print("2.maximum:",max(lst))
print("3.minimum:",min(lst))
print("4.Sorted:", sorted(lst))
print("5.Count of First Element:",lst.count(lst[1]))
print("6.Index of first element:",lst.index(lst[1]))
print("7.Reversed:",lst[::-1])
print("8.Append Element:",lst+["New"])
print("9.Repetition:",lst*2)
print("10.List type:",type(lst))

```

```

List: ['2', '3', '4', '3', '2', '4', '5', '5', '4']
1.length: 9
2.maximum: 5
3.minimum: 2
4.Sorted: ['2', '2', '3', '3', '4', '4', '4', '5', '5']
5.Count of First Element: 2
6.Index of first element: 1
7.Reversed: ['4', '5', '5', '4', '2', '3', '4', '3', '2']
8.Append Element: ['2', '3', '4', '3', '2', '4', '5', '5', '4', 'New']
9.Repetition: ['2', '3', '4', '3', '2', '4', '5', '5', '4', '2', '3', '4', '3', '2', '4', '5', '5', '4']
10.List type: <class 'list'>

```

In [91]: `#dictionary`

```

d= dict(item.split(":")for item in input("Enter key:value pairs seperated by s
print("dictionary:",d)
print("1.Length:",len(d))
print("2.Keys:",d.keys())
print("3.Values:",d.values())

```

```

print("4.Items:",d.items())
print("5.Copy:",d.copy())
print("6.Pop first key:",d.pop(list(d.keys())[0]))
print("7.Type:",type(d))

```

```

dictionary: {'1': 'ved', '2': 'bhavi', '3': 'tanvi', '4': 'snehal', '5': 'mit'}
1.Length: 5
2.Keys: dict_keys(['1', '2', '3', '4', '5'])
3.Values: dict_values(['ved', 'bhavi', 'tanvi', 'snehal', 'mit'])
4.Items: dict_items([('1', 'ved'), ('2', 'bhavi'), ('3', 'tanvi'), ('4', 'snehal'), ('5', 'mit')])
5.Copy: {'1': 'ved', '2': 'bhavi', '3': 'tanvi', '4': 'snehal', '5': 'mit'}
6.Pop first key: ved
7.Type: <class 'dict'>

```

In [113]: *#function , Function with parameter*

```

def add(a,b):
    result = a+b
    return result

def subtract(a,b):
    result = a-b
    return result

def multiply(a,b):
    result = a*b
    return result

def divide(a,b):
    result = a/b
    return result

x=int(input("Enter first number:"))
y=int(input("enter second number:"))

sum=add(x,y)
print("Sum = ",sum)

subtraction=subtract(x,y)
print("Subtraction = ",subtraction)

multi=multiply(x,y)
print("Multiplication = ",multi)

divide=divide(x,y)
print("Divide = ",divide)

```

```

Sum = 10
Subtraction = 0
Multiplication = 25
Divide = 1.0

```

In [9]: *#built in functions*

```

num= int(input("Enter a number:"))
lst= list(map(int,input("enter elements seperated by space : ").split()))
print("List",lst)

print("1.absolute value:",abs(num))
print("2.power:",pow(num,2))
print("3.length:",len(lst))
print("4.maximum:",max(lst))
print("5.minimum:",min(lst))
print("6.Sum:",sum(lst))
print("7.sorted:",sorted(lst))
print("8.round:",round(num/3,2))
print("9.type:",type(lst))
print("10.range:",list(range(num)))

```

```

List [3, 4, 5, 6, 7, 3, 4]
1.absolute value: 5
2.power: 25
3.length: 7
4.maximum: 7
5.minimum: 3
6.Sum: 32
7.sorted: [3, 3, 4, 4, 5, 6, 7]
8.round: 1.67
9.type: <class 'list'>
10.range: [0, 1, 2, 3, 4]

```

In [7]: *#Anonymous function - Lambda,filter,map*

```

lst= list(map(int,input("enter elements seperated by space : ").split()))

even_numbers= list(filter(lambda x:x % 2 == 0,lst))

square_lst=list(map(lambda x:x*x,even_numbers))

print("Original list:",lst)
print("Even Numbers:",even_numbers)
print("Squared Even Numbers:",square_lst)

```

```

Original list: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Even Numbers: [2, 4, 6, 8, 10]
Squared Even Numbers: [4, 16, 36, 64, 100]

```

In [9]: *# maps, list, comprehension, dictionaries*

```

numbers= list(map(int,input("enter elements seperated by space : ").split()))

squares=list(map(lambda x:x*x,numbers))

even_numbers= [x for x in numbers if x % 2 == 0]

num_square_dict={x:x*x for x in numbers}

```

```
print("numbers:",numbers)
print("squares(map):",squares)
print("Even Numbers(list comprehension):",even_numbers)
print("Dictionary (number: square):",num_square_dict)
```

```
numbers: [1, 2, 3, 4, 5]
squares(map): [1, 4, 9, 16, 25]
Even Numbers(list comprehension): [2, 4]
Dictionary (number: square): {1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

In [11]: *#class and object*

```
class student:

    def get_data(self,name,roll):
        self.name=name
        self.roll=roll

    def show_data(self):
        print("Name:",self.name)
        print("Roll No:",self.roll)

s1 = student()

s1.get_data("vedanti",101)
s1.show_data()
```

```
Name: vedanti
Roll No: 101
```

In [19]: *#array*

```
a=[10,20,30,40,50]
print("Array:",a)

a.append(60)
print("after append:",a)

a.insert(2,25)
print("after insert:",a)

a.remove(30)
print("after remove:",a)

a.pop()
print("after pop:",a)

a.sort()
print("after sort:",a)
```

```
Array: [10, 20, 30, 40, 50]
after append: [10, 20, 30, 40, 50, 60]
after insert: [10, 20, 25, 30, 40, 50, 60]
after remove: [10, 20, 25, 40, 50, 60]
after pop: [10, 20, 25, 40, 50]
after sort: [10, 20, 25, 40, 50]
```

In [25]: *#decision and looping statement*

```
for letter in 'vedanti':
    if(letter=='n'):
        print("letter is present")
        break
    else:
        print("current letter",letter)

var =10
while var>0:
    print("current variable value:",var)
    var=var-1
    if var == 5:
        break
```

```
current letter v
current letter e
current letter d
current letter a
letter is present
current variable value: 10
current variable value: 9
current variable value: 8
current variable value: 7
current variable value: 6
```

In [11]: *#writing text files,append ,read*

```
f=open("C:/Users/ACER/OneDrive/Desktop/hello ved.txt","w")
f.write("file is open and write")
f.close

f=open("C:/Users/ACER/OneDrive/Desktop/hello ved.txt","a")
f.write("file is appended")
f.close()

f=open("C:/Users/ACER/OneDrive\Desktop\hello ved.txt","r")
content= f.read()
print("content of file")
print(content)
f.close()
```

```
content of file
file is open and writefile is appended
```

```

<>:3: SyntaxWarning: invalid escape sequence '\0'
<>:11: SyntaxWarning: invalid escape sequence '\D'
<>:3: SyntaxWarning: invalid escape sequence '\0'
<>:11: SyntaxWarning: invalid escape sequence '\D'
C:\Users\ACER\AppData\Local\Temp\ipykernel_11132\2962704259.py:3: SyntaxWarning: invalid escape sequence '\0'
    f=open("C:/Users/ACER/OneDrive/Desktop/hello ved.txt","w")
C:\Users\ACER\AppData\Local\Temp\ipykernel_11132\2962704259.py:11: SyntaxWarning: invalid escape sequence '\D'
    f=open("C:/Users/ACER/OneDrive\Desktop\hello ved.txt","r")

```

In [1]: *# Gui widgets*

```

import tkinter as tk
from tkinter import ttk

root = tk.Tk()
root.title("GUI Widget Example")

root.geometry("400x300")

label = tk.Label(root, text="Hello, Tkinter!", font=("Arial", 16), fg="blue")
label.pack(pady=10)

entry = tk.Entry(root, width=30, font=("Arial", 12))
entry.insert(0, "Enter text here...")
entry.pack(pady=10)

def on_button_click():
    label.config(text=f"Hello, {entry.get()}!")
button = tk.Button(root, text="Click Me!", command=on_button_click, font=("Arial", 12))
button.pack(pady=10)

combo = ttk.Combobox(root, values=["Option 1", "Option 2", "Option 3"],
font=("Arial", 12))
combo.current(0)
combo.pack(pady=10)

check_var = tk.IntVar()
checkbox = tk.Checkbutton(root, text="Check me!", variable=check_var, font=("Arial", 12))
checkbox.pack(pady=10)

label.config(fg="green")
entry.config(width=40)

quit_button = tk.Button(root, text="Quit", command=root.quit, font=("Arial", 12), fg="red")
quit_button.pack(pady=20)

root.mainloop()

```



In [3]: *#creating layouts, packing order,controlling widgets*

```
import tkinter as tk

root = tk.Tk()
root.title("Layout and Packing Order Example")
root.geometry("400x400")

frame1 = tk.Frame(root, bg="lightblue", height=100, width=400)
frame1.pack(fill="x", padx=10, pady=10)

frame2 = tk.Frame(root, bg="lightgreen", height=100, width=400)
frame2.pack(fill="both", expand=True, padx=10, pady=10)

frame3 = tk.Frame(root, bg="lightyellow", height=100, width=400)
frame3.pack(fill="x", padx=10, pady=10)

label1 = tk.Label(frame1, text="Top Frame (Light Blue)", font=("Arial", 14))
label1.pack(pady=20)

label2 = tk.Label(frame2, text="Middle Frame (Light Green)", font=("Arial", 14))
label2.pack(side="left", padx=20, pady=20)
button1 = tk.Button(frame2, text="Button 1", font=("Arial", 12))
button1.pack(side="left", padx=20)
button2 = tk.Button(frame2, text="Button 2", font=("Arial", 12))
button2.pack(side="right", padx=20)

label3 = tk.Label(frame3, text="Bottom Frame (Light Yellow)", font=("Arial", 14))
label3.pack(pady=20)
quit_button = tk.Button(frame3, text="Quit", font=("Arial", 12), command=root.destroy, fg="red")
quit_button.pack(pady=10)

root.mainloop()
```

In [ ]: